From Retrospective Value to Prospective Design

Valuing Forecast Improvements Under a Nonstationary Climate

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Roadmap

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I: What We Know

The Retrospective Result

We quantified the social value of augmented observations (AO) for eight landfalling hurricanes (2017–2020), combining data-denial experiments (Ditchek et al. 2023) with a structural cost model (Molina et al. 2024).

Three key estimates:

1. **Loss curvature:** $\widehat{\beta}_2^w = 5.49$ million per unit increase in squared demeaned wind error, for hurricane-force counties.
2. **Precision gain from AO:** $\widehat{\delta} = 0.26$ m/s reduction in forecast standard deviation at the 72-hour lead time.
3. **Per-county value:** $\approx$ \$25.77 million per affected county per storm.

Benefit-to-cost ratio $\approx$ 50:1 against flight operations.

The Valuation Formula

Hurricane costs depend on two features of the forecast environment: the *level* of the storm (mean intensity μ) and the *precision* of the forecast (error variance σ^2). The expected cost can be written as:

$$\mathbb{E}[C] = f(\mu) + \hat{\beta}_2^w \cdot \sigma^2$$

where $f(\mu)$ captures the direct cost of storm intensity and $\hat{\beta}_2^w \cdot \sigma^2$ captures the cost of forecast uncertainty. The social value of a precision improvement that reduces error variance from σ_0^2 to σ_1^2 is:

$$V = \hat{\beta}_2^w \left(\sigma_0^2 - \sigma_1^2\right)$$

Key property: Because hurricane damages are *convex* in wind speed, both the level μ and the scale σ^2 matter. A higher μ increases the baseline cost and can amplify the marginal cost of errors. A higher σ^2 raises the baseline from which improvements must operate.

The Stationarity Assumption

All retrospective estimates draw from the 2005–2022 climate.

The valuation formula treats four objects as fixed:

1. $\widehat{\beta}_2^w$: the loss curvature linking forecast errors to costs.
2. μ: the location of the intensity distribution (mean wind speed for affected counties).
3. σ_0^2: the baseline forecast error variance (status quo technology).
4. $\Delta\sigma^2$: the variance reduction achievable by a given observational investment.

Under a changing climate, some (or all) of these are stationary. Climate might shift both the *location* μ (higher peak intensities, more Cat 4–5 events) and the *scale* σ^2 (greater forecast uncertainty from expanded atmospheric variability).

II: Why Nonstationarity Matters

Two Channels of Climate Influence

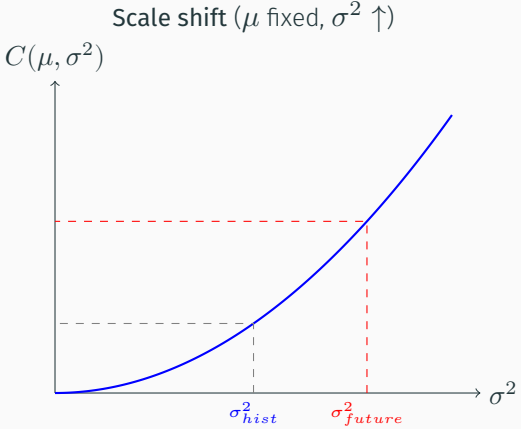
Climate change might change the value of forecast precision through two channels:

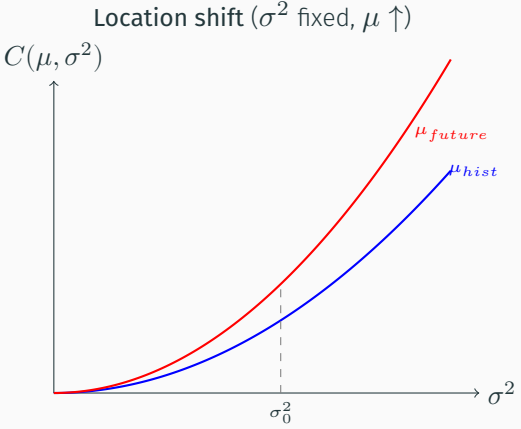
1. Location shift ($\mu \uparrow$). The intensity distribution shifts rightward. Projections might indicate a higher proportion of Category 4–5 events and increased peak wind speeds. More county-storms cross the hurricane-force threshold ($>$33 m/s), enlarging the set over which $\widehat{\beta}_2^w$ applies. Because damages are convex in wind speed, a higher μ also amplifies the marginal cost of forecast errors at any given σ^2.

2. Scale shift ($\sigma^2 \uparrow$). Greater variability, rapid intensification episodes, and thermodynamic environments increases the irreducible component of forecast error variance. The forecast models must operate on a distributional support that is expanding.

Both channels raise the social value of a given precision improvement $\Delta\sigma^2$.

Convexity: Both Channels Amplify Value





Left: Same $\Delta\sigma^2$, but higher baseline σ^2 means steeper cost curve $\Rightarrow$ larger V. **Right:** Same σ^2 and $\Delta\sigma^2$, but higher μ rotates the cost curve upward $\Rightarrow$ larger V.

III: A Framework for Prospective Valuation

The Forecast Error Model

For location i, lead time s periods ahead from time t:

$$y_{i,t+s} = \hat{g}(X_t) + \epsilon_{i,t+s}, \quad \epsilon_{i,t+s} \sim F_s(0,\ \sigma_s^2)$$

where X_t is the vector of current and lagged atmospheric states and $\hat{g}(\cdot)$ is the forecast model's best prediction. Under climate change, the distribution of outcomes $y_{i,t+s}$ shifts in both location and scale:

- **Location:** The mean of the intensity distribution μ shifts rightward (higher V_{max}, more rapid intensification episodes, etc). This increases the probability of extreme realizations and the baseline cost against which errors are evaluated.
- **Scale:** The conditional forecast error variance σ_s^2 increases as the atmospheric state space expands beyond the historical support on which $\hat{g}(\cdot)$ was estimated.

Both shifts raise the social value of any given precision improvement $\Delta\sigma^2$.

A Prospective Valuation Approach

Define the value of a forecast improvement under climate scenario λ, which determines both the location and scale of the intensity distribution:

$$V(\lambda) = \beta(\mu(\lambda)) \cdot \Delta\sigma^2(\lambda)$$

where:

- $\mu(\lambda)$: the location of the intensity distribution under scenario λ. A higher μ increases baseline damages and can amplify the loss curvature β.
- $\sigma_0^2(\lambda)$: the *status quo* forecast error variance under scenario λ.
- $\Delta\sigma^2(\lambda) = \sigma_0^2(\lambda) - \sigma_1^2(\lambda)$: the variance reduction from the improved technology.

Climate change raises V through both parameters: μ and σ_0^2. These effects compound: the same technology is worth more when storms are both stronger *and* harder to predict.

A Semi-Parametric Simulation

Step 1: Empirical baseline. Estimate the joint distribution of (μ, σ_s^2) from the historical sample of data-denial experiments (e.g., Ditchek et al. 2023), stratified by lead time and intensity category. This is the known object.

Step 2: Climate perturbation. Shift (μ, σ^2) (e.g., under CMIP6 scenarios, SSP2-4.5 vs. SSP5-8.5). The location shift draws on projected changes in TC intensity distributions (e.g., Knutson et al., 2020). The scale shift reflects increased forecast error variance from expanded atmospheric variability.

Step 3: Propagation. For each draw from the perturbed distribution, compute $V(\lambda) = \beta(\mu(\lambda)) \cdot \Delta\sigma^2(\lambda)$ and aggregate across county-storms to obtain scenario-specific total values.

Step 2: Two Perturbation Channels

Location channel ($\mu \uparrow$). Use projected rightward shift of the TC intensity distribution to perturb μ.

- Literature suggests a modest increase in mean peak wind (about 3% under 2°C warming) but a substantially heavier right tail, with 10% more Category 4–5 storms.
- A higher μ both enlarges the set of hurricane-force county-storms and steepens the loss function $\beta(\mu)$. **Scale channel ($\sigma^2 \uparrow$).** Scale σ_s^2 by a multiplicative factor $\kappa > 1$ reflecting the increase in irreducible forecast uncertainty.
    - This channel captures the effect of expanding atmospheric variability on model skill and is the more speculative of the two.

A key question for this audience: What is the best empirical basis for calibrating the scale factor κ?

An analogous framework applies to track uncertainty. Define location-level exposure as:

$$p_{i,t+s} = \Pr(\text{location } i \text{ within strike zone at } t+s \mid X_t)$$

This can be implemented as:

- A binary indicator: whether the TC centroid passes through location i.
- A kernel-weighted function: exposure that decays with distance from the forecast center, e.g., $K\left(\frac{\|r_i - r_{center}\|}{R}\right)$ for some radius R.

A Simple Illustration

A Simple Illustration: Setup

We compute $V(\mu, \sigma^2) = \beta(\mu) \times \Delta\sigma^2$ over a grid of plausible (μ, σ_0^2) pairs.

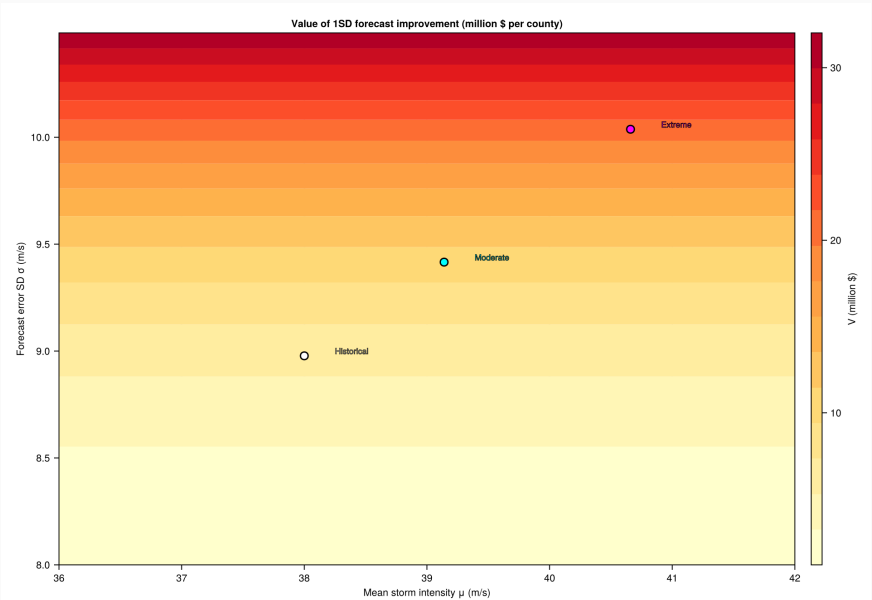
Calibration from the retrospective analysis:

- $\widehat{\beta}_2^w = 5.49$ M per (m/s)^2, at a historical mean intensity $\bar{\mu} = 38$ m/s.
- $\Delta\sigma^2 = 1$ (m/s)^2, hypothetical variance reduction from augmented observations.
- $\beta(\mu)$ is increasing in μ above the hurricane-force threshold (33 m/s), reflecting the convexity of damages in wind speed: $\beta(\mu) = \widehat{\beta}_2^w \left(\frac{\mu - 33}{\bar{\mu} - 33}\right)^3$.

Tentative climate scenarios shift (μ, σ_0^2) jointly:

- Moderate μ increases 3%, σ^2 scales by $\kappa = 1.10$.
- Extreme μ increases 7%, σ^2 scales by $\kappa = 1.25$.

A Simple Illustration: Value Surface $V(\mu, \sigma^2)$



Open Questions

Open Questions for the Community

1. **Calibrating the scale channel.** What is the best empirical or model-based strategy for projecting how σ_s^2 scales with changes in the large-scale environment?

2. **Structural stability of precision gains.** Should we expect the physical improvement from a given observational investment (the $\Delta\sigma^2$) to remain constant as (μ, σ^2) shift? Or does data assimilation performance itself degrade (or improve) in a warmer climate?

3. **Multi-hazard extension.** The current framework is wind-only. Precipitation and storm surge channels likely amplify the total value. How should we prioritize these extensions, and what data would be needed?

We welcome any and all feedback on the scale-shift calibration!